Exam intro

Exam Introduction to Data Science 1MS041.

Information

The exam consists of three main topics

1. Singular Value Decomposition
2. Classification
3. Markov chains

Each topic has roughly 9 smaller parts. Each small subpart can be answered independently of each other. For each such topic you have a left pane with the problem setup and the right pane contains the particular question.

To pass the exam you need 20 points out of 40.

Allowed aids

At the computer you are writing the exam you have access to:

- The github course website with all the couse notebooks and the lecture notes
- Numpy documentation
- Github
- Online compilation environments: myCompiler, Programiz, Online Python, GDB Online, try-jupyter
- Python documentation
- The Wasserman book

Links to these aids are available as tabs at the bottom of Inspera

There are many questions that require you do do calculations, some require a bit of hand work on paper to get right and then you can calculate in one of these online environments. There is one problem that requires you to implement your own code in which case you will need to copy it over to the answer box in Inspera. Select the code with CTRL-A, copy with CTRL-C and paste it in the Inspera answer box using CTRL-V. You can undo with CTRL-Z.

While you write your exam your answers are saved every 15s. Copy often from the online programming environments to not risk loosing code. You cannot open the exam once you have handed it in.

Good luck!

Consider the matrix

$$M = \begin{pmatrix} 1 & 0 \\ 2 & 2 \\ 0 & 1 \end{pmatrix}.$$

Parts (a) and (b) concern M; part (c) moves from M to principal component analysis of a real dataset.

Work the singular value decomposition $M = UDV^T$ **by hand**. A useful first step is to form $M^T M$, whose eigenvalues are the squared singular values and whose eigenvectors are the right singular vectors.

Where a singular vector is asked for, use the convention that its **first non-zero component is positive**, and order the singular values from largest to smallest.

Matrix norm: For a matrix $A = (a_{ij})$, the Frobenius norm is $\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2}$

1 Rank of M

Give the rank of M.: [ ] (2) .

Maximum marks: 0.5

2 Largest singular value

Give the largest singular value σ_1 of M. [ ] (2.995 - 3.005) .

Maximum marks: 0.5

3 Smallest singular value

Give the smallest singular value σ_2 of M. [ ] (0.995 - 1.005) .

Maximum marks: 0.5

4 Right singular vectors

In the decomposition $M = UDV^T$, the columns of V are the right singular vectors, ordered so that the largest singular value comes first. Each candidate below is described by its two columns, v_1 then v_2.

A: $v_1 = \frac{1}{\sqrt{2}}(1,\ 1)^T, v_2 = \frac{1}{\sqrt{2}}(1,\ -1)^T$

B: $v_1 = (1,\ 1)^T, v_2 = (1,\ -1)^T$

C: $v_1 = \frac{1}{\sqrt{2}}(1,\ -1)^T, v_2 = \frac{1}{\sqrt{2}}(1,\ 1)^T$

D: $v_1 = \frac{1}{\sqrt{5}}(2,\ 1)^T, v_2 = \frac{1}{\sqrt{5}}(1,\ -2)^T$

Which one is a valid V?

Select one alternative:

- B
- A ✔
- D
- C

Maximum marks: 1

5 Left singular vector

Which vector is u_1, the left singular vector belonging to the **largest** singular value?

Select one alternative:

- $u_1 = (1, 4, 1)^T$
- $u_1 = \frac{1}{\sqrt{2}} (1, 0, -1)^T$
- $u_1 = \frac{1}{3} (1, 4, 1)^T$
- $u_1 = \frac{1}{\sqrt{18}} (1, 4, 1)^T$ ✔

Maximum marks: 1

6 Code bug

A student checks their decomposition in NumPy:

```
import numpy as np
M = np.array([[1, 0], [2, 2], [0, 1]], dtype=float)
U, s, Vt = np.linalg.svd(M)
D = np.diag(s)
print(U @ D @ Vt)
```

and gets the response

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[1], line 9
      5 U, s, Vt = np.linalg.svd(M)
      7 D = np.diag(s)
----> 9 print(U @ D @ Vt)
ValueError: matmul: Input operand 1 has a mismatch in its core dimension 0,
with gufunc signature (n?,k),(k,m?)->(n?,m?) (size 2 is different from 3)
```

Select one alternative:

- `Vt` holds the right singular vectors as rows, so line 5 should multiply by `Vt.T`.
- `M` must be square before `np.linalg.svd` can be used; a 3×2 matrix has to be padded with a zero column first.
- `s` is returned as a one-dimensional array of length 2, so `np.diag(s)` is 2×2 when a 3×2 matrix of singular values is needed.
- `np.linalg.svd` returns the full decomposition by default, so `U` is 3×3 while `D` is 2×2. The product `U @ D` is not conformable.

Maximum marks: 1.5

7 Construct M1

Using the first singular component, form
$M_1 = \sigma_1 u_1 v_1^T$
which matrix is M_1

A: $\begin{pmatrix} 1/2 & 1/2 \\ 2 & 2 \\ 1/2 & 1/2 \end{pmatrix}$
B: $\begin{pmatrix} 1 & 1 \\ 4 & 4 \\ 1 & 1 \end{pmatrix}$
C: $\begin{pmatrix} 1/6 & 1/6 \\ 2/3 & 2/3 \\ 1/6 & 1/6 \end{pmatrix}$
D: $\begin{pmatrix} 1/2 & -1/2 \\ 0 & 0 \\ -1/2 & 1/2 \end{pmatrix}$

Select one alternative:

- D
- A ✔
- C
- B

Maximum marks: 2

8 Frobenius

Let
$M_1 = \sigma_1 u_1 v_1^T$
be the rank-1 approximation obtained by keeping only the first singular component of M. Give the value of the fraction

$\frac{\|M_1\|_F^2}{\|M\|_F^2} =$ [ ] (0.89 - 0.91)

Maximum marks: 1

9 # PCA

In part 1 the singular values of M were computed directly, without subtracting the column means. Principal component analysis, by contrast, always begins by centring the data column-wise.

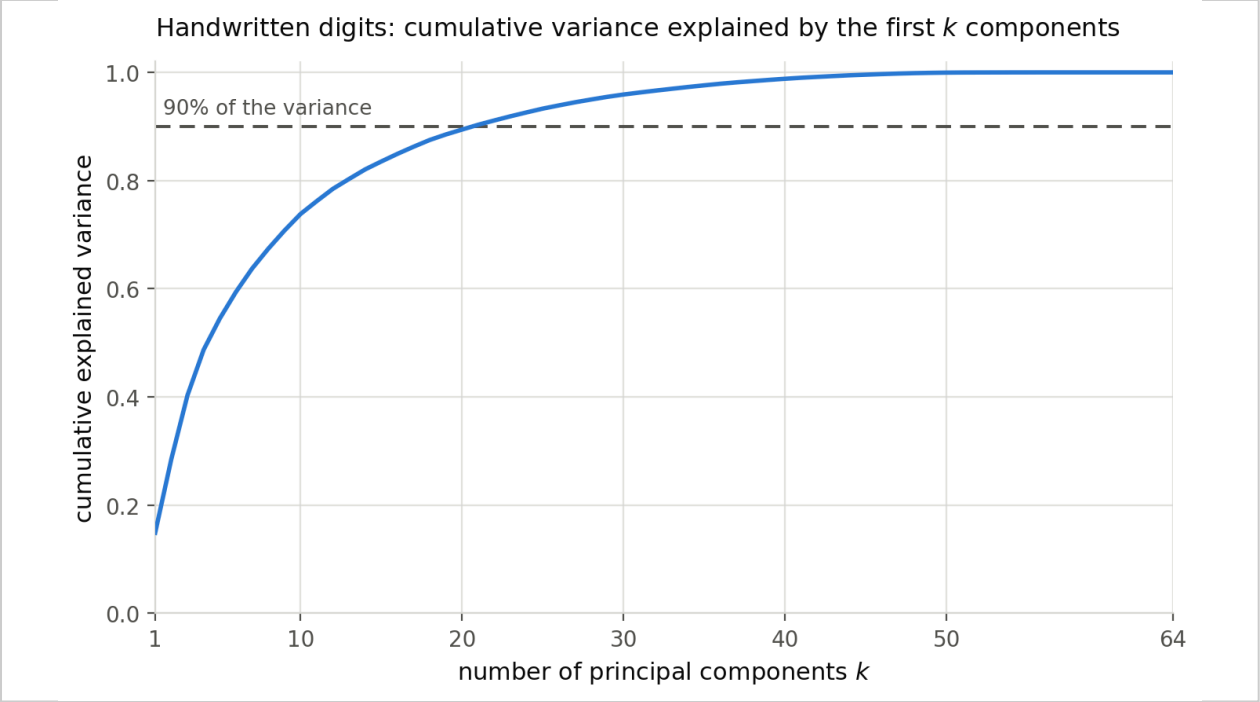
Why does PCA require this step?

Select one alternative:

- Centring is needed to make the principal components orthogonal.
- Centring makes every feature have the same variance, preventing large-scale features from dominating.
- Centring puts every feature on a comparable scale, which the SVD requires in order to be numerically stable.
- Principal components describe variation around the mean. Without centring, the leading directions may instead reflect the position of the data relative to the origin, rather tha ✔ variation around the mean.
- Without centring the matrix may be rank-deficient, and the SVD is only defined for full-rank matrices.

Maximum marks: 2

10 SVD components



The figure shows the cumulative explained variance of the 8 by 8 handwritten digit images, which have 64 pixel features, after centring the data. The dashed line marks 90% of the variance.

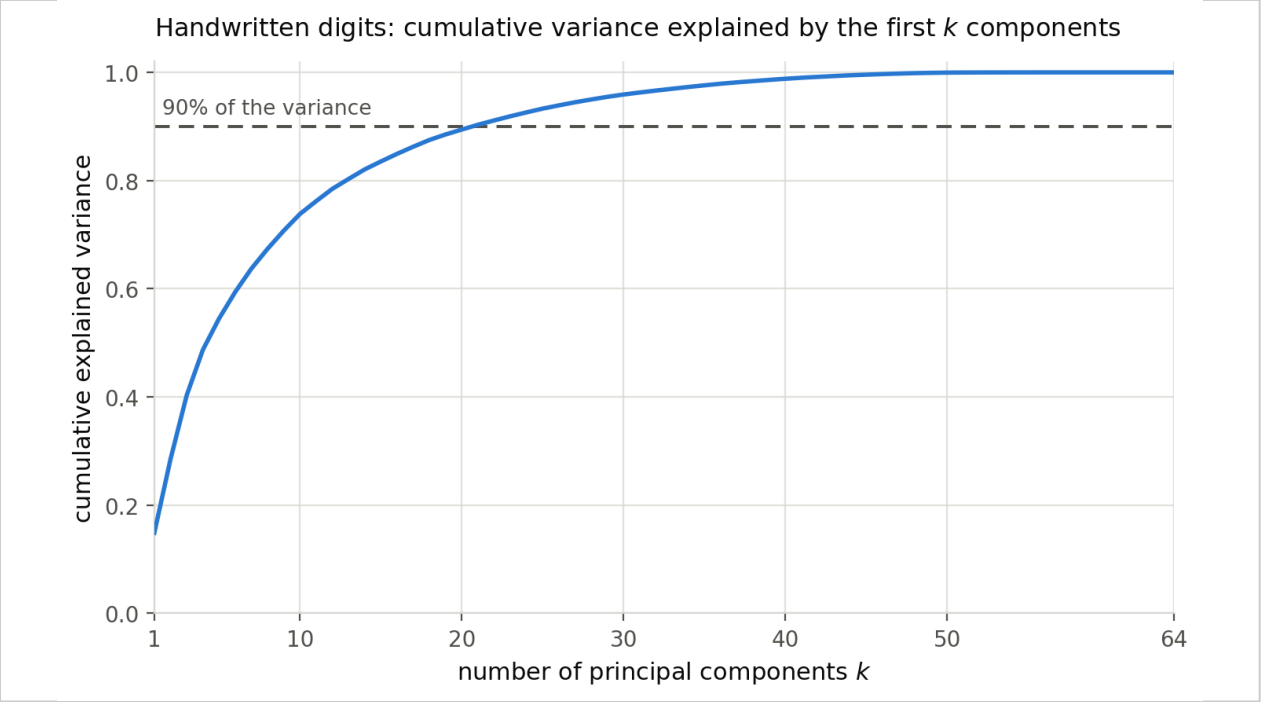
What is the smallest number of principal components needed to explain at least 90% of the variance?

Select one alternative:

- 13
- 5
- 29
- 21 ✔

Maximum marks: 1

11 SVD components2



The figure shows the cumulative explained variance of the 8 by 8 handwritten digit images, which have 64 pixel features, after centring the data. The dashed line marks 90% of the variance.

Which statement best describes what it tells you about the digit images?
Select one alternative:

- About a third of the 64 dimensions suffice to capture 90% of the variation, so the pixel features are highly redundant and the images lie close to a much lower-dimensiona ✔ subspace of the original 64-dimensional pixel space.
- Each of the 64 components explains roughly the same share of the variance, about 1.6% each.
- The first 21 of the 64 pixels carry 90% of the information in each image.
- A classifier using the first 21 components will be about 90% accurate.

Maximum marks: 1

12 What is being computed?

Consider the following code:

```
X = X - X.mean(axis=0)
U, s, Vt = np.linalg.svd(X, full_matrices=False)

k = 5
q = np.sum(s[:k]**2) / np.sum(s**2)
```

What does q represent?

Select one alternative:

- The classification accuracy obtained after reducing the data to 5 dimensions.
- The fraction of the original features that are retained.
- The fraction of the variance in the centred data explained by the first 5 principal components. ✔
- The fraction of observations that can be reconstructed exactly using 5 components.

Maximum marks: 2

A spam filter assigns each SMS message a score in [0,1] and predicts **spam** when the score is at least a decision threshold t. The filter is evaluated on a held-out test set of **200 messages, of which 100 are spam and 100 are ham**.

At the default threshold $t = 0.4$ the results are

Confusion matrix

	Predicted ham	**Predicted spam**
True ham	84	16
True spam	36	64

Throughout, *precision* and *recall* refer to the **spam** class.

13 Accuracy

From the confusion matrix (on the panel to the left), give the **accuracy** of the filter on the test set. Put the answer here [ ] (0.73 - 0.76)

Maximum marks: 1

14 Precision

Give the **precision** for the spam class, that is the fraction of the messages predicted spam that really are spam. Put your answer here: [ ] (0.79 - 0.81)

Maximum marks: 1

15 Recall

At threshold t=0.4, which statement about the spam-class recall is correct? We define recall as the fraction of the true spam messages that the filter finds.

Select one alternative:

- The recall is 64/(64+36)=0.64. If the threshold is increased, recall can only stay the same or decrease, because some messages currently predicted as spam may instead be predicted as ham. ✔
- The recall is (84+64)/200=0.74. Increasing the threshold does not affect recall.
- The recall is 64/(64+16)=0.80. If the threshold is increased, recall will increase because predictions become more selective.
- The recall is 36/(64+36)=0.36. Increasing the threshold can only decrease the number of false negatives.

Maximum marks: 1

16 # Hoeffding

At threshold t=0.4, the filter has empirical accuracy 0.74 on n=200 test messages.

For a quantity bounded in [0,1], Hoeffding's inequality gives a 95% confidence interval with half-width

$$\varepsilon = \sqrt{\frac{\log(2/\alpha)}{2n}}, \alpha = 0.05$$

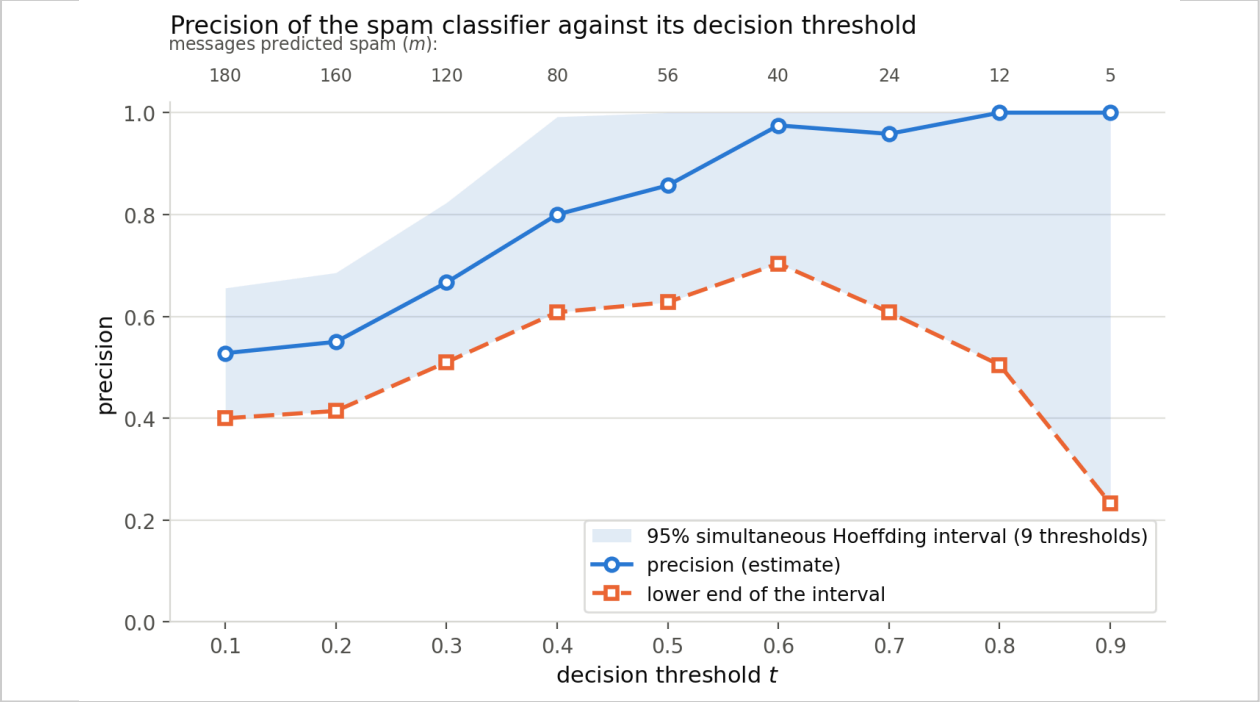
A manager will deploy the filter only if the test data support the claim that its **true accuracy is at least 0.65** with 95% confidence. What conclusion should you draw?

Select one alternative:

- Yes. A 95% confidence interval means that at least 95% of future messages will be classified correctly.
- Yes. The observed accuracy is 0.74, which is larger than 0.65.
- No. A confidence interval cannot be used to assess whether an unknown true accuracy is larger than a fixed value such as 0.65.
- No. The Hoeffding half-width is approximately 0.096, so the lower endpoint is 0.74-0.096=0.644, which is below 0.65. ✔

Maximum marks: 2

17 Threshold guarantee



The figure shows the estimated precision of the spam filter as the decision threshold t varies, together with 95% confidence intervals that hold **simultaneously for all nine thresholds**.
You must choose a threshold using these data. You want to make the largest defensible statement of the form

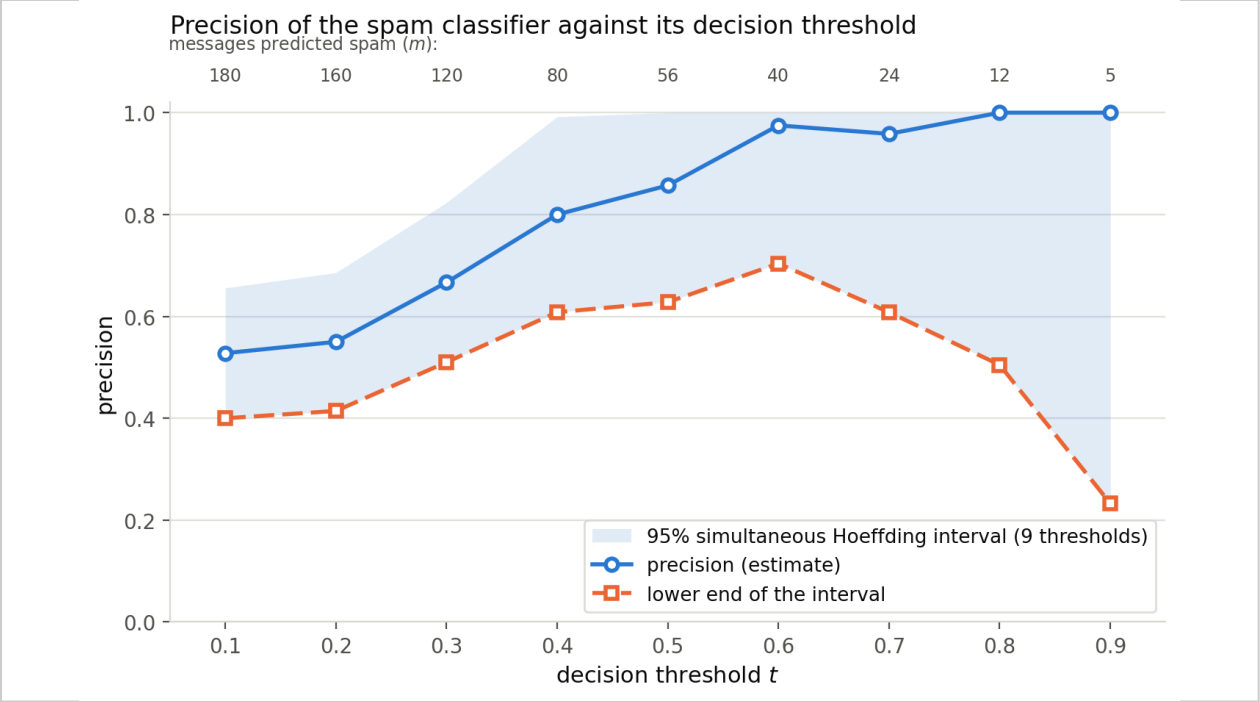
“the true precision is at least p.”

at the 95% confidence level.
Which threshold should you choose?
Select one alternative:

- t=0.9, where the precision estimate is 1.00
- t=0.8, the largest threshold whose precision is 1.00
- t=0.4, the default threshold, since it classifies the most messages confidently
- t=0.6 ✔

Maximum marks: 2

18 Increasing width



As the decision threshold increases, the confidence intervals for precision become much wider, even though the estimated precision increases.

Which explanation best accounts for this?

Select one alternative:

- Hoeffding intervals become wider when the estimated probability is close to 1.
- At higher thresholds, fewer messages are predicted as spam. Precision is therefore estimated from a smaller sample, so the Hoeffding half-width increases. ✔
- Increasing the threshold also increases the confidence level, which produces wider intervals.
- A higher threshold makes the classifier intrinsically less reliable, so the uncertainty in its precision must increase.

Maximum marks: 1

19 Choice of C

You want confidence intervals for precision at all nine thresholds to hold simultaneously with probability at least 0.95.

Using the union bound, what value of C should be used in

$$\varepsilon = \sqrt{\frac{2C/\alpha}{2m}}$$

to calculate the confidence interval half width?

Select one alternative:

- C=9, because nine confidence intervals must hold simultaneously. ✔
- C=0.05, because the significance level is 5%.
- C=200, because there are 200 test messages.
- C=1, because each individual interval is computed from one threshold at a time.

Maximum marks: 1

20 Sample size

A future evaluation of the same classifier again obtains empirical accuracy 0.74.
You want the 95% Hoeffding lower confidence bound for the true accuracy to be at least 0.65.
What is the **smallest test-set size n** that would make this possible?
Use

$$\varepsilon = \sqrt{\frac{\log(2/\alpha)}{2n}}, \alpha = 0.05$$

Select one alternative:

- 456
- 228 ✔
- 114
- 41

Maximum marks: 3

21 Score function

A spam classifier assigns each message a score in [0,1] and predicts spam when `score >= t`. The list `y_true` contains `1` for spam and `0` for ham.

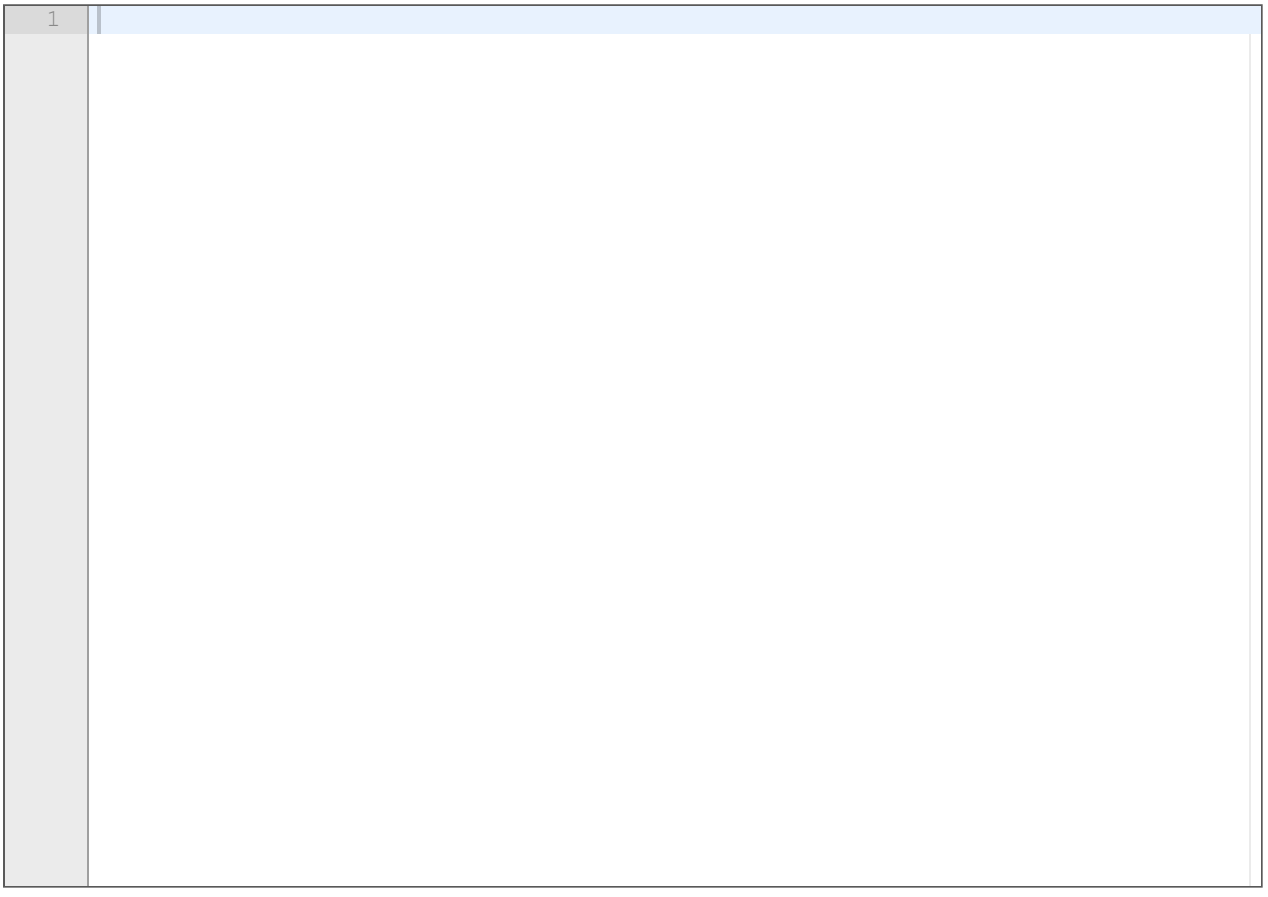
Write a NumPy-based Python function. Convert the inputs to NumPy arrays and use vectorized operations for thresholding and counting; pandas and scikit-learn are not needed.

```python
def evaluate(scores, y_true, t):
# return (accuracy, precision, recall)
```

Use the definitions from Questions 13–15. Include a call that evaluates

```python
scores = np.array([0.10, 0.35, 0.40, 0.55, 0.80, 0.90, 0.20, 0.45])
y_true = np.array([0, 0, 1, 0, 1, 1, 1, 0])
t = 0.40
```

Include `import numpy as np`. The code is preferably checked in one of the online Python compilers listed under references.
Write your answer in the box below. Changes are saved automatically.



Maximum marks: 2

A warehouse tracks packages moving between three zones, in this fixed order throughout the whole problem:

$$\mathbf{Dock}, \quad \mathbf{Sort}, \quad \mathbf{Dispatch}.$$

For a particular package, let X_t denote the zone containing the package after t transitions. We model $(X_t)_{t\geq 0}$ as a homogeneous Markov chain on the three states above. Writing N_{ij} for the number of observed transitions from zone i to zone j, the logged counts are (row is from and column is to)

	Dock	Sort	Dispatch
Dock	40	160	0
Sort	30	120	150
Dispatch	60	120	60

No package was ever observed going directly from Dock to Dispatch.

22 MLE

Using the observed transition counts in the left panel, estimate
$P_{\mathrm{Dock,Sort}} = \Pr(X_{t+1} = \mathrm{Sort} \mid X_t = \mathrm{Dock}).$

Using maximum likelihood. Give your answer as a decimal here [ ] (0.795 - 0.805)

Maximum marks: 1

23 Two step

use the estimated transition matrix

$$\hat{P} = \begin{pmatrix} 0.20 & 0.80 & 0 \\ 0.10 & 0.40 & 0.50 \\ 0.25 & 0.50 & 0.25 \end{pmatrix},$$

with rows and columns ordered as Dock, Sort, Dispatch.

Suppose that X_0=Dock. What is the probability that the package is in Dispatch after exactly two transitions, that is,
$\Pr(X_2 = \mathrm{Dispatch} \mid X_0 = \mathrm{Dock})$

Put your answer here: [ ] (0.395 - 0.405)

Maximum marks: 1

24 Two step 2

A student wants to compute the matrix of two-step transition probabilities from the estimated transition matrix P:

They run the following code:

```
P = np.array([ [0.20, 0.80, 0.00], [0.10, 0.40, 0.50], [0.25, 0.50, 0.25] ])
P2 = P**2
```

They expect P2[i,j] to equal
$\mathbb{P}(X_2 = j \mid X_0 = i)$
but the result is incorrect.

What is the problem?
Select one alternative:

- The transition matrix must first be transposed, so the correct calculation is `P.T @ P.T`.
- The exponent should be written `P^2`; in NumPy `^` denotes matrix powers.
- Two-step transition probabilities are obtained by multiplying every entry of `P` by 2, so the correct calculation is `2 * P`.
- `P**2` squares each entry of `P` separately. Two-step transition probabilities require the matrix product, so one should use `P @ P` (or `np.linalg.matrix_power(P, 2)`). ✔

Maximum marks: 2

25 Which matrix?

$$A = \begin{pmatrix} 0.1 & 0.4 & 0.2 & 0.2 & 0.1 \\ 0.2 & 0.1 & 0.3 & 0.3 & 0.1 \\ 0.1 & 0.2 & 0.2 & 0.3 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.1 & 0.3 \\ 0.3 & 0.1 & 0.2 & 0.2 & 0.2 \end{pmatrix}, B = \begin{pmatrix} 0 & 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.4 & 0.6 \\ 0 & 0 & 0 & 0.7 & 0.3 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$C = \begin{pmatrix} 0.3 & 0.3 & 0.2 & 0.1 & 0.1 \\ 0.2 & 0.3 & 0.3 & 0.1 & 0.1 \\ 0.1 & 0.2 & 0.4 & 0.2 & 0.1 \\ 0 & 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0 & 0.4 & 0.6 \end{pmatrix}, D = \begin{pmatrix} 0 & 0 & 0.5 & 0.3 & 0.2 \\ 0 & 0 & 0.2 & 0.5 & 0.3 \\ 0.6 & 0.4 & 0 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 & 0 \\ 0.3 & 0.7 & 0 & 0 & 0 \end{pmatrix}$$

Match the matrix with the description of its properties

	A	C	B	D
Irreducible and aperiodic	✔			
Not irreducible		✔		
Irreducible with period 3			✔	
Irreducible with period 2				✔

Maximum marks: 3

26 Stationary Sort

For the estimated transition matrix

$$\hat{P} = \begin{pmatrix} 0.20 & 0.80 & 0 \\ 0.10 & 0.40 & 0.50 \\ 0.25 & 0.50 & 0.25 \end{pmatrix},$$

let

$\pi = (\pi_{\text{Dock}}, \pi_{\text{Sort}}, \pi_{\text{Dispatch}})$

be a stationary distribution, so that

$\pi\hat{P} = \pi, \qquad \pi_{\text{Dock}} + \pi_{\text{Sort}} + \pi_{\text{Dispatch}} = 1.$

Give $\pi_{\text{Sort}} =$ [ ] (0.495 - 0.505)

Maximum marks: 1

27 Stationary

Let π be a stationary distribution of the estimated transition matrix $\hat{P}$. Which statement correctly describes the meaning of π?

Select one alternative:

- If the chain is irreducible, the entries of π give the long-run fractions of time spent in the different states. If the chain is also aperiodic, the distribution of X_t converges to π ✔ regardless of the initial state.
- The entries of π are the probabilities of moving from the current state to each of the three states in the next transition.
- A stationary distribution describes only the distribution of X_0, it does not contain information about the long-run behaviour of the chain.
- For every Markov chain, the distribution of X_t converges to π regardless of the initial state as soon as a stationary distribution exists.

Maximum marks: 1

28 Hitting time

A package starts in Dock.

Define

$$T = \min\{t \geq 0 : X_t = \text{Dispatch}\},$$

the number of transitions until the package first reaches Dispatch.

Using

$$\hat{P} = \begin{pmatrix} 0.20 & 0.80 & 0 \\ 0.10 & 0.40 & 0.50 \\ 0.25 & 0.50 & 0.25 \end{pmatrix},$$

give

$\mathbb{E}[T \mid X_0 = \text{Dock}] =$ [ ] (3.495 - 3.505)

Maximum marks: 3